

BFCI at AraFinNLP2024: Support Vector Machines for Arabic Financial Text Classification

Nsrin Ashraf¹

Hamada Nayel¹

Mohammed Aldawsari³

Shashirkha H. L.²

Tarek Elshishtawy¹

¹Faculty of Computers and Artificial Intelligence, Benha University, Egypt

²Department of Computer Science, Mangalore University, India

³Faculty of Engineering, Prince Sattam bin Abdulaziz University, Saudi Arabia

Correspondence: nisrien.ashraf19@fci.bu.edu.eg

Abstract

In this paper, a description of the system submitted by BFCI team to the AraFinNLP 2024 shared task has been introduced. Our team participated in the first subtask, which aims at detecting the customer intents of cross-dialectal Arabic queries in the banking domain. Our system follows the common pipeline of text classification models using primary classification algorithms integrated with basic vectorization approach for feature extraction. Multi-layer Perceptron, Stochastic Gradient Descent and Support Vector Machines algorithms have been implemented and support vector machines outperformed all other algorithms with an f-score of 49%. Our submission's result is appropriate compared to the simplicity of the proposed model's structure.

1 Introduction

Natural Language Processing (NLP) is massively growing area of research. It has been implemented in general domain as well as specific domains such as biomedical (Shashirekha and Nayel, 2016), scientific, and financial (Patil et al., 2023) domains. A system that summarizes financial documents automatically using NLP approaches has been developed by Patil et. al (Patil et al., 2023).

Processing Arabic texts is extensively growing due to the increasing amount of texts written in Arabic. The complex morphological structure is the major challenge that faces researchers in Arabic Natural Language Processing (NLP) (AbuElAtta et al., 2023). Financial Arabic NLP is introduced as a result of the exponential growth of Middle Eastern stock markets with abroad range of diverse sectors. This rise, which is occurring in multiple regions, is indicative of the dynamic financial landscape in the region and is drawing interest and investment from around the world. NLP technologies are becoming essential as these markets develop in order to handle regional linguistic quirks and

serve the international financial community that is involved in these exchanges. The efficient analysis and interpretation of financial data depends on the development of Arabic NLP tools in the banking industry (Zmandar et al., 2023, 2021). The Arabic Financial NLP (AraFinNLP) shared task aims at analysing financial texts written in Arabic (Malaysha et al., 2024).

AraFinNLP shared task aims at improving Financial Arabic NLP. It consists of two subtasks namely; Multi-dialect Intent Detection, and Cross-dialect Translation and Intent Preservation, in the banking domain. These subtasks are crucial for interpreting and managing the diverse and complex banking data prevalent in Arabic-speaking regions. Detecting intent in financial communications, particularly in bots, across various Arabic dialects, can enhance customer service, and automate query handling. This ensures inclusivity and efficiency in catering to a linguistically diverse customer base. The Dialectal Translation aspect is particularly significant given the linguistic diversity in the Arab world. It ensures that NLP models are not only accurate when dealing with Modern Standard Arabic (MSA) but also effective across diverse Arabic dialects. This advancement will open up new applications in areas like automated customer support, real-time financial news analysis, and enhanced accessibility for diverse Arabic-speaking populations, making financial services more inclusive and efficient.

ArBanking77 (Jarrar et al., 2023), a translated version of English Banking77 dataset (Casanueva et al., 2020) into MSA and Palestinian Arabic has been used for AraFinNLP 2024. Task organizers shared the dataset across all participants, providing training, development and blind test sets. Participants have to develop their models using train and development sets, and the performance of their models will be evaluated using the blind test set.

Our team participated on the first subtask: Multi-

dialect Intent Detection in the banking domain. The subtask automates the classification process of customer intents given the query in different Arabic dialects. We proposed a machine learning-based model for the first subtask, by integrating a basic vectorization approach and support vector machines algorithm. Such approach has been applied for various arabic NLP tasks (Ashraf et al., 2022).

2 Dataset

The dataset given for the shared task was obtained from English Banking77 Dataset (Casaneve et al., 2020) consists of 13,083 costumer service queries divided into 77 classes according to the customer intents. ArBanking77 dataset (Jarrar et al., 2023) contains MSA and Palestinian Arabic, For the proposed Model Our team participated in Subtask-1 using the MSA dataset, as the statistics of dataset is shown in Table 1.

Train	Validation	Test
10733	1230	11721

Table 1: ArBanking77 Dataset Statistics.

3 Proposed Model

In this section, a detailed description of the proposed model including the main structure, text representation, algorithms, experimental setup and evaluation metrics has been presented. As shown in Figure 1, the first phase of the proposed model is data distribution for MSA data presented in Table 1. The second phase is feature engineering, which aims at converting the text into numerical values efficiently. The third phase is training the model, or implementation of the classification algorithm. The final phase is evaluation, which is performed in the organizers side. The following subsections give more details for each phase separately.

3.1 Feature Engineering

Feature engineering phase proposes a conversion of the row text data into numerical values, as most algorithms accept numerical input of a fixed size rather than text data of different sizes. A collective technique employs a document-term vector, in which each document is encoded as a discrete vector that adds occurrences for each word in the vocabulary it includes (Rameshbhai and Paulose, 2019). Term Frequency/Term Document Frequency, *TF/IDF*, is one of simple and efficient

text vectorization approaches where it takes into account the importance of each term in the document. It can handle high-dimensional data by reducing the feature space to the most important terms. In this work, we employed *TF-IDF* with *n*-gram model using *uni*-gram ($n = 1$) and *bi*-gram ($n = 2$). *N*-gram model captures more complex relationships between words, by considering both single words and pair of words, other parameters set CountVectorizer CV=5, MaxDF=1.0. This method can lead to more accurate text analysis (Nayel, 2020).

3.2 Training the Model

In this phase, the resulted numerical features are used as input to the classification algorithm for training. Three classification algorithms have been implemented to broaden the experiments namely; Multi-Layer Perceptron (MLP), Support Vector Machine (SVM) and Stochastic Gradient Descent (SGD). These algorithms have been successfully implemented for different Arabic NLP tasks such as dialect identification (Sobhy et al., 2022) and offensive language identification (Nayel, 2020)

- **SVM** is one of the most effective ML text classifiers in multi-label text classification which has the ability to handle high-dimensional data with a relatively minimal number of training samples.
- **MLP** is an artificial neural network that is widely utilized in classification and regression problems. In the context of multi-label text categorization, MLPs can be utilized to learn the complicated relationships between input text and output labels.
- **SGD** is more efficient in Multi-label text classification compared to other optimization algorithms which has several parameters that can be tuned to improve performance (Diab, 2019).

3.3 Evaluation

F1-score, also known as F-measure, is a harmonic mean of Precision (P) and Recall (R). It has been used to evaluate the model performance. Equation 1, shows that calculation of F1-score using P and R .

$$F1 - score = \frac{2 * P * R}{P + R} \quad (1)$$

F1-score is commonly used for multi-class classification problems

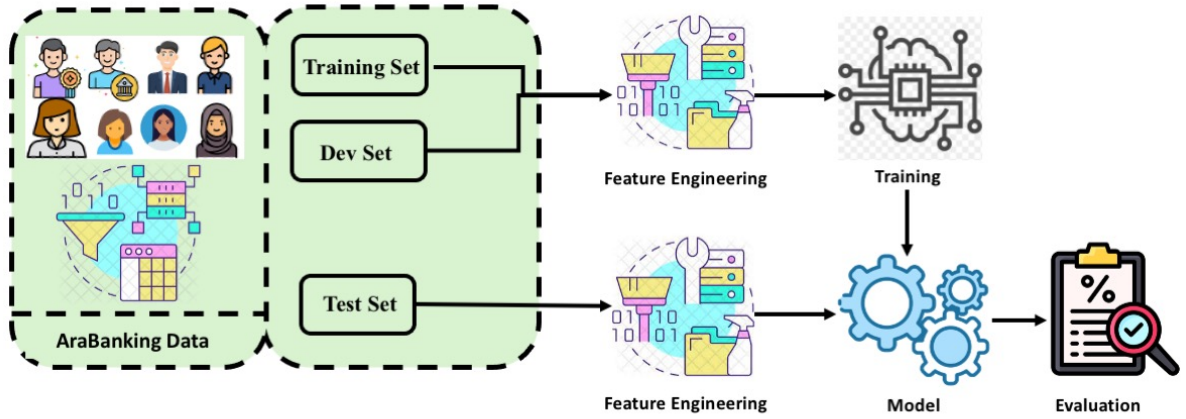


Figure 1: The general structure of the proposed model

4 Experimental Setup and Results

We used Jupyter notebook¹ for GPU support with python programming language and Sklearn (Pedregosa et al., 2011) library was utilized for machine learning classifiers. To achieve the same results, we set random state of the random seed package at a specific value for each classifier. The hyper-parameters of the implemented classifiers are as follows:

- MLP has been trained using the following back-propagation variations: hidden layers = 20, solver = 'adam', activation function = logistic, maximum iteration = 500 and random state = 42
- SGD has been trained using 'log' loss function, penalty = 12, maximum iteration = 1000 and random state = 10
- SVM uses 'linear' kernel, regularization = 'L1' and random state = 42

Table 2 shows F1-score reported on development phase for the three models. It is clear that SVM outperforms both SGD and MLP. We submitted the output of SVM to the shared task, and the result of SVM for the test set is 49.10% f1-score.

¹<https://jupyter.org/>

Classifier	F1-Score
MLP	84.46%
SVM	85.12%
SGD	74.65%

Table 2: ArBanking77 F1-Score for Development Phase

5 Discussion

The proposed model follows the basic pipeline of NLP-based model and no external resources have been employed. The issue of overfitting has been raised in the proposed model, where f1-scores are disparately resulted on development set and the blind test set. The SVM classifier outperform MLP and SGD classifiers in the presence of overfitting during the training phase due to its built-in regularization term that stimulate the model to find a simpler decision boundary. Using TF-IDF vectorization approach and machine learning algorithms resulted 49% Micro F1-Score achieving rank 11 in the shared task.

6 Future Work

This work can be enhanced by employing additional classification algorithms, where we implemented only three basic algorithms. In addition, deep neural networks can be implemented for the given task. On the other hand, using advanced text vectorization approach can lead to improvement of the performance. Finally, large language models (LLMs) can be customised to this task.

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